# AI TRACEFINDER

## Document Source Identification using Scanner Pattern

**Final Report**

Infosys Springboard Internship

Submitted by:

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## 1.Introduction

With the increasing use of scanned documents for official and legal purposes, ensuring the authenticity and origin of scanned images has become an important challenge. Each scanner device introduces unique noise patterns and artifacts during the scanning process. These scanner-specific fingerprints can be utilized to identify the source scanner used to scan a document.

The goal of **AI TraceFinder** is to identify the scanner device (brand/model) used to produce a scanned document by analyzing scanner pattern traces. This project is highly useful in **digital forensics**, **document verification**, and **legal evidence authentication**.

## 2. Problem Statement

The main aim of this project is to identify the scanner device used to scan a document by analyzing unique noise patterns or artifacts left by the scanner hardware. Each scanner introduces device-specific sensor noise that can be learned and classified using machine learning and deep learning approaches.

## 3.Objectives

The key objectives of AI TraceFinder are:

- Collect and organize scanned datasets from multiple scanners.
- Preprocess and normalize scanned images for analysis.
- Extract meaningful scanner-specific features (PRNU, texture, FFT, noise).
- Train baseline machine learning models for classification.
- Develop robust PRNU-correlation based scanner identification.
- Train and evaluate a deep learning CNN model on PRNU-based images.
- Deploy a simple Streamlit UI for real-time scanner identification and logging.

## 4. Use Cases

**4.1 Digital Forensics**
Helps identify which scanner was used to forge or duplicate legal documents.
**4.2 Document Authentication**
Helps differentiate scans from authorized and unauthorized sources.
**4.3 Legal Evidence Verification**
Ensures scanned copies submitted for legal purposes came from approved scanners.

## 5. Dataset Description

The project uses scanned document images categorized into three main dataset types:

- **Flatfield**: blank scans capturing strong scanner noise patterns
- **Official**: real-world official documents
- **Wikipedia**: diverse layouts and document structures

Each dataset is organized into subfolders based on scanner model (such as **Canon, Epson, HP**).
Folder name acts as the label for each scanner class.

## 6. System Architecture

The complete workflow of TraceFinder follows the below stages:

1. Data Collection & Labeling
2. Preprocessing (resize, grayscale, normalization)
3. Feature Engineering (FFT, LBP, noise, PRNU statistics)
4. Baseline Model Training (LogReg, SVM, Random Forest)
5. PRNU Correlation Matching
6. Deep Learning CNN Training + Explainability (Grad-CAM)
7. Deployment (Streamlit UI + logging + CSV download)

## 7. Methodology and Implementation

### 7.1 Milestone-1: Dataset Collection & Preprocessing

Week-1 and Week-2 focused on building a clean dataset foundation.

Key tasks:

- Dataset collected and organized by scanner model
- A labels.csv file was automatically generated mapping image paths to scanner labels
- Images were preprocessed using:
  - Grayscale conversion
  - Resizing to 512 × 512
  - Removal of junk files like macOS hidden metadata
- Processed images stored separately in a Processed/ folder maintaining the original structure

✅ Outcome: A clean and consistent dataset ready for feature extraction and modeling.

### 7.2 Milestone-2: Feature Engineering & Baseline Modeling

Milestone-2 focused on scanner fingerprint analysis using hand-crafted features and classical ML models.

Extracted Features:

- Intensity features: Mean, standard deviation
- FFT feature: frequency domain magnitude
- Texture features: LBP mean and standard deviation
- Noise variance feature: residual noise variance
- PRNU statistical features: PRNU mean, energy, variance

Baseline Models Trained:

- Logistic Regression (~74%)
- SVM (~76%)
- Random Forest (~75%)

Although these models performed reasonably, accuracy was limited due to loss of spatial noise patterns.

### 7.3 PRNU-Correlation Method (Proposed)

To improve accuracy, a PRNU correlation-based approach was implemented.

Steps:

- Extract PRNU residuals from images using Gaussian filtering
- Generate scanner-level fingerprints by averaging residuals of multiple images
- Compare test image PRNU residual against stored fingerprints using normalized cross-correlation

✅ Result: 96.97% Accuracy, significantly higher than baseline ML models.

### 7.4 Milestone-3: Deep Learning Model + Explainability

In Milestone-3, a CNN model was trained using PRNU-based images.

CNN Architecture Highlights:

- Convolution layers for feature extraction
- Batch Normalization for stability
- MaxPooling for downsampling
- Fully Connected layers
- Softmax output for multi-class classification

Model Performance:

- Training Accuracy: 97.28%
- Validation Accuracy: 96.34%
- Test Accuracy: 93.08%

### 7.5 Explainability using Grad-CAM

Grad-CAM was applied to visualize which regions contributed to predictions.

**Observation:**

Heatmaps mainly focused on noise-like patterns across the image instead of document text, confirming that the CNN relies on scanner-specific PRNU fingerprints rather than content.

✅ This improves forensic reliability and transparency.

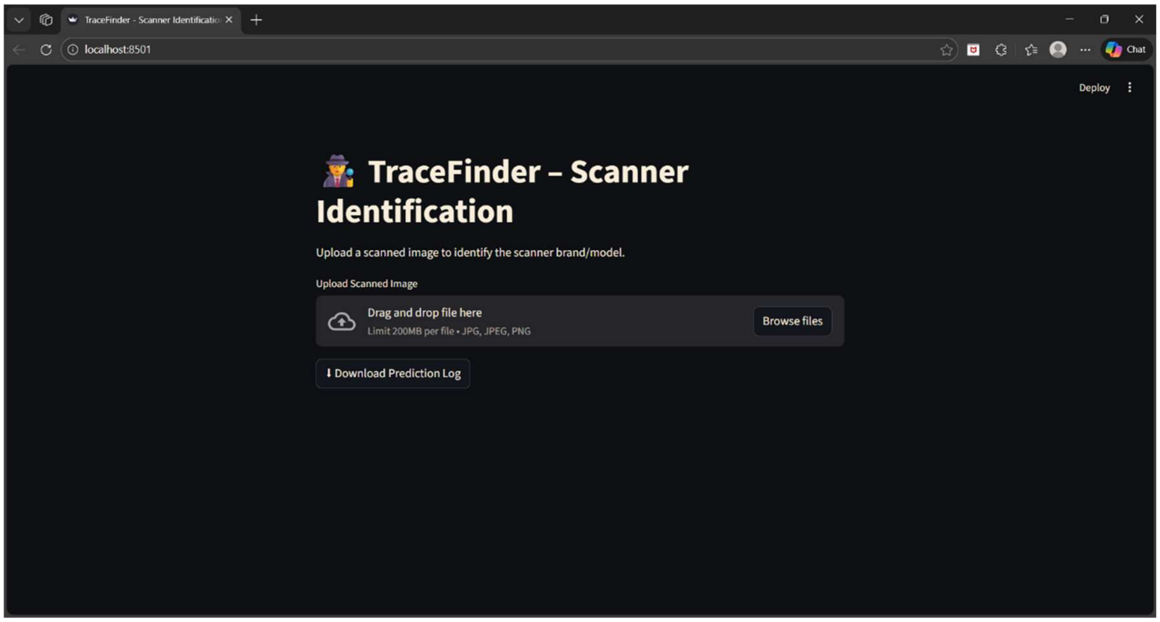
## 8. Milestone-4: Deployment using Streamlit

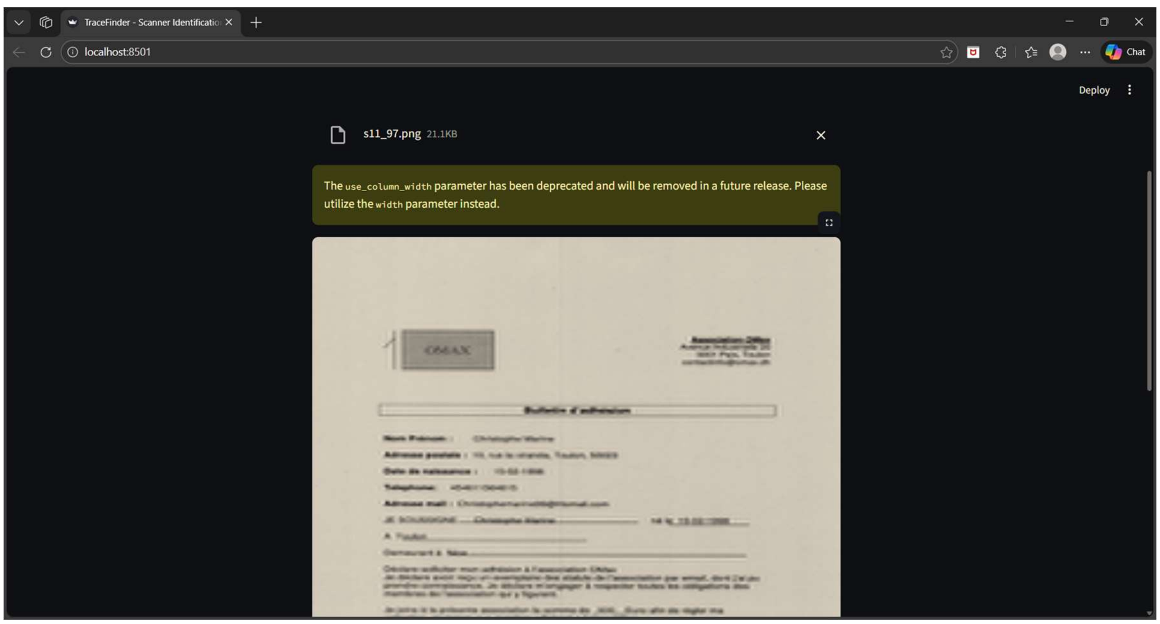
In Milestone-4, the trained CNN model was deployed using Streamlit.

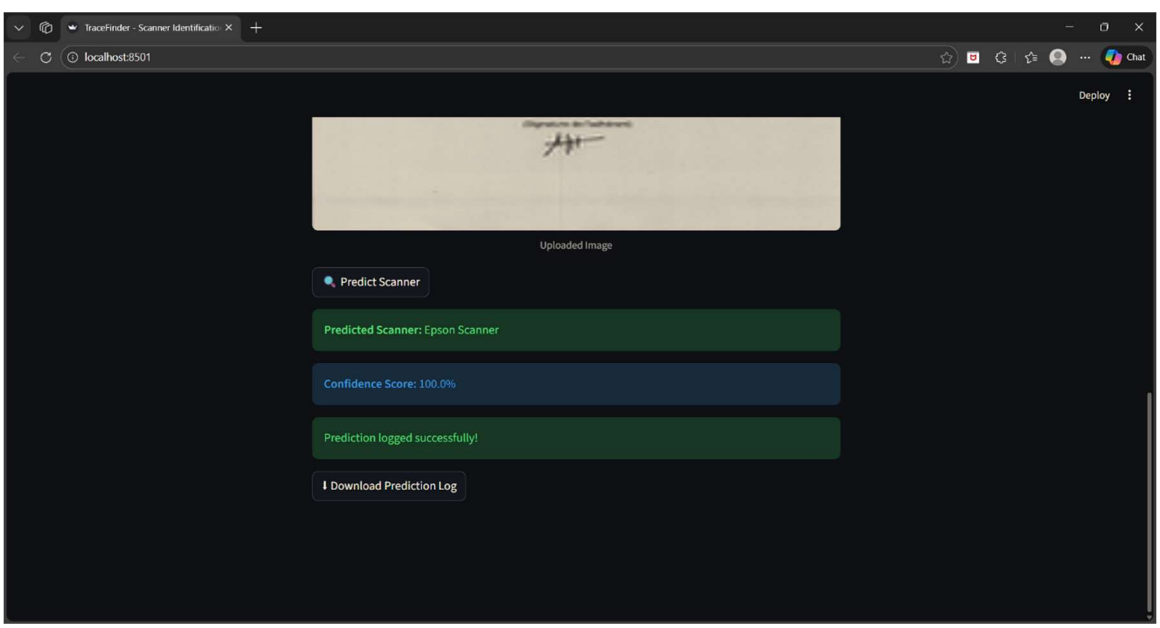
**Features Implemented:**

✅ Upload scanned image (PNG/JPG/JPEG)
✅ Predict scanner model/brand
✅ Display confidence score
✅ Log predictions into predictions.csv
✅ Download prediction logs as CSV

The deployment makes the project usable in real-time for practical verification tasks.


TraceFinder - Scanner Identificatio
localhost:8501
Chat
Deploy
TraceFinder – Scanner Identification
Upload a scanned image to identify the scanner brand/model.
Upload Scanned Image
Drag and drop file here
Limit 200MB per file • JPG, JPEG, PNG
Browse files
Download Prediction Log


TraceFinder - Scanner Identificatio
localhost:8501
Chat
Deploy
s11_97.png 21.1KB
The use_column_width parameter has been deprecated and will be removed in a future release. Please utilize the width parameter instead.


TraceFinder - Scanner Identificatio
localhost:8501
Chat
Deploy
Uploaded Image
Predict Scanner
Predicted Scanner: Epson Scanner
Confidence Score: 100.0%
Prediction logged successfully!
Download Prediction Log

## 9. Results Summary

| **Method** | **Accuracy** |
|---|---|
| Logistic Regression | ~74% |
| SVM | ~76% |
| Random Forest | ~75% |
| **PRNU Correlation (Best)** | **96.97%** |
| CNN Deep Learning Model (Test) | 93.08% |

The PRNU correlation approach achieved the best accuracy and proved highly effective for scanner source identification.

## 10. Conclusion

AI TraceFinder successfully demonstrates a complete forensic scanner identification system using scanner noise patterns. The project achieved high accuracy using PRNU correlation and deep learning CNN models. Deployment using Streamlit provides a user-friendly real-time application with prediction logging and result download.

This project can strongly support forensic investigations and document verification processes.

## 11. Future Scope

Future improvements for TraceFinder include:

- Increasing dataset size with more scanner models
- Improving CNN robustness with better PRNU augmentation strategies
- Adding mobile/web deployment for real-world usage
- Integrating automatic PRNU extraction and correlation into real-time pipeline
- Adding advanced explainability methods like SHAP for deeper insights